

A Kernel-Surjectivity Obstruction for a Proposed Minimum-Cost Extension of Popular Maximum Matchings

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Abstract

This note checks whether a proposed proof of a polynomial-time algorithm for minimum-cost popular maximum matchings in hospitals/residents instances can be repaired by optimizing cost over the matroid-kernel construction of Csáji, Király, Takazawa, and Yokoi. I do not obtain a solution to the Kavitha–Makino incomplete-preference problem. The positive part that survives is the standard encoding of hospitals/residents matchings as common independent sets of two ordered partition matroids, together with the known theorem that one can find some popular maximum-cardinality solution in the corresponding two-sided ordered-matroid model. The secondary cost step is false: a popular maximum matching need not be the projection of any kernel in the duplicated critical instance. I give a five-edge counterexample in which the unique minimum-cost popular maximum matching is not kernel-liftable, while every kernel projects to a more expensive popular maximum matching. Thus the proposed proof is not a solution; the result here is partial progress in the negative direction, identifying a precise obstruction that any successful repair must overcome.

1 Status and definitions

The status of this note is deliberately modest. It does not solve the minimum-cost popular matching problem for hospitals/residents instances with incomplete preferences. It proves that one natural attempted repair of the matroid-intersection approach fails. The obstruction is not merely a gap in exposition: the central “cost-surjectivity” assertion needed for secondary optimization is false.

Let $G = (R \cup H, E)$ be a hospitals/residents instance. Every resident has capacity one and a strict order over its acceptable hospitals. Hospital h has capacity q_h and a strict order over its acceptable residents. A matching $M \subseteq E$ satisfies

$$|M \cap E(r)| \leq 1 \quad (r \in R), \quad |M \cap E(h)| \leq q_h \quad (h \in H).$$

For two matchings M, N , a resident votes $+1, 0, -1$ according as its assignment in M is better than, equal to, or worse than its assignment in N , with being unmatched last. A hospital first deletes common residents from $M(h)$ and $N(h)$, then adds dummy residents to the smaller side so the two residual sets have equal cardinality. Dummies are ranked below all real residents. The hospital vote is the minimum, over bijections between the two residual sets, of the sum of the pairwise votes in favor of M . Write

$$\Delta(M, N)$$

for the total resident-plus-hospital vote. A matching is called *popular maximum* here if it has maximum cardinality and satisfies

$$\Delta(M, N) \geq 0 \quad \text{for every maximum-cardinality matching } N.$$

This is weaker than ordinary popularity, where N ranges over all matchings.

2 The sound first-order reduction

The following standard observation is sound and useful, but by itself it has no secondary-cost content.

Proposition 1 (Partition-matroid encoding). *Let $S = E$. Define two partition matroids*

$$\mathcal{M}_R = \bigoplus_{r \in R} U_{1, E(r)}, \quad \mathcal{M}_H = \bigoplus_{h \in H} U_{q_h, E(h)}.$$

Order the summand $E(r)$ according to resident r 's preferences and $E(h)$ according to hospital h 's preferences. Then hospitals/residents matchings are exactly the common independent sets of $\mathcal{M}_R$ and $\mathcal{M}_H$. Moreover, the two-sided ordered-matroid vote agrees with the vote Δ above.

Proof. The common-independent-set constraints are exactly the resident and hospital capacity constraints. On a resident summand $U_{1, E(r)}$, a comparison has at most one real exchange, so it is exactly the resident's vote. On a hospital summand $U_{q_h, E(h)}$, the feasible exchange pairings are precisely the bijections between the residual sets after common residents have been removed, with dummy elements used to equalize cardinalities. Summing over all summands gives Δ . $\square$

A theorem of Csáji–Király–Takazawa–Yokoi says, in particular, that in the two-sided ordered-matroid model with total orders, a popular maximum-weight common independent set exists and can be found in polynomial time. Applying that theorem with the primary weight $w(e) = 1$ gives some popular maximum-cardinality hospitals/residents matching. This recovers existence and first-order optimization. It does not imply that a minimum-cost popular maximum matching can be obtained by minimizing cost over the kernels used inside that construction.

3 The attempted cost extension and the missing surjectivity

The proposed proof tried to do the following. First, take an optimal chain-supported dual solution for the primary matroid-intersection LP. Restrict to tight elements and impose the corresponding critical equalities. Then replace the critical instance by a duplicated matroid-kernel instance. Finally, give every copy of an original element the same secondary cost and compute a minimum-cost kernel.

The projection direction of the known critical-kernel construction says that a kernel in the duplicated instance projects to a popular critical common independent set. For minimum-cost optimization, however, one also needs the converse:

Every popular critical common independent set should have a kernel lift with the same projection, and hence the same cost.

This converse is the false step. The next sections give a small counterexample. The example is useful because it is not a pathological failure of notation: it shows the precise conceptual problem. A kernel is a local stability object, while popularity permits non-local cancellations of votes along alternating structures.

4 A five-edge hospitals/residents instance

Let

$$R = \{r_0, r_1, r_2\}, \quad H = \{h_0, h_1\}, \quad q_{h_0} = q_{h_1} = 2.$$

The acceptable edges are denoted

$$a = r_0 h_1, \quad b = r_1 h_0, \quad c = r_1 h_1, \quad d = r_2 h_0, \quad e = r_2 h_1.$$

Preferences are

$$r_0 : a, \quad r_1 : c \succ b, \quad r_2 : e \succ d,$$

while the hospital preferences are

$$h_0 : d \succ b, \quad h_1 : e \succ a \succ c.$$

There are exactly three maximum-cardinality matchings:

$$A = \{a, b, d\}, \quad B = \{a, c, d\}, \quad C = \{a, b, e\}.$$

Indeed, a maximum matching must match all three residents. It must use a for r_0 . The four possible choices for r_1 and r_2 are bd, be, cd, ce , and the choice ce overfills h_1 .

Lemma 2. *The majority margins among maximum-cardinality matchings are*

$\Delta(\text{row}, \text{column})$	A	B	C
A	0	-1	-1
B	1	0	0
C	1	0	0

Consequently, B and C are popular maximum matchings, while A is not.

Proof. We compute the nontrivial comparisons. For B against A , resident r_1 improves from b to c , giving $+1$, and the other residents are unchanged. At h_0 , after deleting the common resident r_2 , matching B compares a dummy to b , giving -1 . At h_1 , after deleting the common resident r_0 , matching B compares c to a dummy, giving $+1$. Thus $\Delta(B, A) = 1$.

For B against C , residents r_1 and r_2 contribute $+1$ and -1 . Hospital h_0 compares d to b and contributes $+1$, while h_1 compares c to e and contributes -1 . Thus $\Delta(B, C) = 0$. The computations for C against A and C against B are symmetric in the same sense and give 1 and 0 , respectively. Antisymmetry of pairwise votes gives the displayed table. $\square$

Now assign secondary costs

$$\text{cost}(e) = 1, \quad \text{cost}(a) = \text{cost}(b) = \text{cost}(c) = \text{cost}(d) = 0.$$

Then B is the unique minimum-cost popular maximum matching: it has cost zero, while the only other popular maximum matching C has cost one.

5 The duplicated critical instance

Consider the primary weight $w \equiv 1$. The maximum cardinality is 3. In the matroid-intersection dual, put dual value 1 on the resident-side set $S = E$ and zero on all other rank constraints. This is feasible and has objective $\text{rank}_{\mathcal{M}_R}(S) = 3$, so it is optimal. All elements are tight. The resulting critical equality is simply $|I| = 3$.

For this dual chain, the duplicated kernel construction has copies

$$f^0, f^1, f^2, f^3 \quad (f \in \{a, b, c, d, e\}).$$

On the resident side, lower levels are preferred; within one level, the original resident preference is used. On the hospital side, higher levels are preferred; within one level, the original hospital preference is used. The projected original edge set of X is denoted by $\pi(X)$.

A set X of copies is a kernel if it is common independent and every outside copy $f^k \notin X$ is dominated on at least one side: on that side, $X + f^k$ is dependent, and every feasible one-element deletion from X that restores independence is preferred to f^k .

Proposition 3. *In the duplicated instance above, every kernel projects to $C = \{a, b, e\}$. Conversely, for each $t \in \{0, 1, 2, 3\}$,*

$$X_t = \{a^t, b^0, e^0\}$$

is a kernel. Hence the kernel projections are exactly the matching C .

Proof. First observe that $\pi(X)$ must be a maximal matching in the original graph. If an acceptable edge $f \notin \pi(X)$ could be added to $\pi(X)$ while preserving all original capacities, then $X + f^0$ would be common independent in the copied instance. The outside copy f^0 would therefore be dominated on neither side, contradicting the kernel property. The maximal matchings in the original graph are

$$A = \{a, b, d\}, \quad B = \{a, c, d\}, \quad C = \{a, b, e\}, \quad D = \{c, e\}.$$

It remains to show that A, B, D have no kernel lifts and that C does.

Suppose first that X projects to A . Write the selected copies as $a^\alpha, b^\beta, d^\delta$. Consider the outside copy e^0 . On the resident side, d^δ is the only possible deletion, but resident r_2 does not prefer d^δ to e^0 : if $\delta = 0$, then r_2 prefers e to d , and if $\delta > 0$, then e^0 is at a better level. On the hospital side, h_1 currently contains only a^α , so adding e^0 does not violate the hospital capacity. Thus e^0 is undominated, a contradiction.

Suppose next that X projects to B , with copies $a^\alpha, c^\beta, d^\delta$. Again consider e^0 . The resident side cannot dominate it, for the same reason as above. Hospital h_1 is full, so hospital-side domination of e^0 would require both a^α and c^β to be preferred by h_1 to e^0 . Since e is h_1 's favorite resident at level 0, this forces $\alpha > 0$ and $\beta > 0$. Now consider the outside copy b^0 . Hospital h_0 currently contains only d^δ , so adding b^0 does not violate the hospital capacity. Therefore b^0 would have to be dominated on the resident side. The only possible deletion is c^β , and resident r_1 prefers c^β to b^0 if and only if $\beta = 0$. This contradicts $\beta > 0$.

Suppose now that X projects to $D = \{c, e\}$, with copies c^β, e^γ . The outside copy a^0 cannot be dominated on the resident side because resident r_0 is unmatched. Hospital h_1 is full, so hospital-side domination of a^0 requires both c^β and e^γ to be preferred to a^0 . Since h_1 ranks a above c at level 0, this requires $\beta > 0$. But now inspect b^0 . Hospital h_0 is empty, so b^0 cannot be hospital-dominated. It must be resident-dominated, and resident r_1 prefers c^β to b^0 if and only if $\beta = 0$. This contradiction excludes D .

Finally fix $t \in \{0, 1, 2, 3\}$ and set $X_t = \{a^t, b^0, e^0\}$. We check domination of every outside copy. Another copy a^k with $k \neq t$ is dominated by the copy constraint: if $k > t$, then a^t is preferred on the resident side, while if $k < t$, then a^t is preferred on the hospital side. Copies b^k and e^k with $k > 0$ are dominated on the resident side by b^0 and e^0 , respectively. For c^k , if $k > 0$, then resident r_1 prefers b^0 to c^k because level 0 is better; if $k = 0$, then h_1 is full and both a^t and e^0 are preferred by h_1 to c^0 . Finally, every copy d^k is dominated on the resident side by e^0 , since either $k > 0$ or $k = 0$ and r_2 prefers e to d . Thus X_t is a kernel. $\square$

Corollary 4. *The duplicated-kernel cost minimization method does not find the minimum-cost popular maximum matching in this instance.*

Proof. By Lemma 2, B is a popular maximum matching of cost zero. By Proposition 3, every kernel projects to C , whose cost is one. Therefore minimizing copied cost over kernels and projecting cannot output the true minimum-cost popular maximum matching. $\square$

The same argument directly refutes the desired cost-surjectivity statement: B is a popular critical common independent set for the chosen critical face, but it has no kernel lift.

6 What is actually obtained

The attempted proof cannot be repaired by merely adding costs to the existing critical-kernel construction. The obstruction is structural: kernel-liftable solutions form a proper subset of the popular critical solutions, and the minimum-cost popular solution may lie outside that subset.

The following statements remain valid.

- (i) Hospitals/residents matchings with incomplete strict lists are common independent sets of two ordered partition matroids, and the ordered-matroid vote agrees with the hospitals/residents vote.
- (ii) The known two-sided ordered-matroid theorem gives a polynomial-time method to find some popular maximum-cardinality matching, where popularity is tested among maximum-cardinality matchings.
- (iii) If one restricts attention to kernel-liftable popular critical solutions, then a minimum-cost kernel gives a minimum-cost member of that restricted subclass. The counterexample shows that this subclass is too small for the desired optimization problem.

Thus this note is partial progress, not a solution. A successful solution to the secondary-cost problem would need either a different characterization of all popular critical solutions or a proof that the Kavitha–Makino objective has additional structure not captured by the kernel projection used here.

References

- [1] G. Csáji, T. Király, K. Takazawa, and Y. Yokoi. Popular maximum-utility matchings with matroid constraints. *arXiv:2407.09798*, 2024.
- [2] T. Kavitha and K. Makino. Min-cost popular matchings in a hospitals/residents instance with complete preferences. *Mathematics of Operations Research*, Articles in Advance, 2025. DOI: 10.1287/moor.2024.0634.